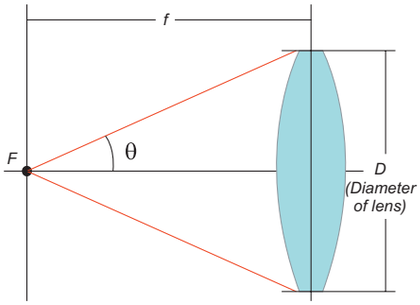
by Ernst Abbe, and is defined by the following expression



Numerical Aperture (NA) =n × sin(θ)

n = the refractive index of the medium between the specimen and objective; θ = half aperture angle or collection angle of the objective. (the maximum half angle of the cone of light that can enter or exit the lens).

Infobits

The smallest cells on the planet are some forms of *Mycoplasma* with dimensions of 0.2 to 0.3 µm, which is within the limit of resolution of light microscopes. Tiny cells that look like dwarf bacteria but are 10 times smaller than *Mycoplasma* and 100 times smaller than the average bacterial cell are called nanobacteria or nanobes (*Greek* nanos means one billionth).

The resolving power of a light microscope depends on the wavelength of light used and the NA of the objective lens.

The numerical aperture of a lens can be increased by

- Increasing the size of the lens opening and/or
- Increasing the refractive index of the material between the lens and the specimen.

The larger the numerical aperture the better the resolving power. It is important to illuminate the specimens properly to have higher resolution. The concave mirror in the microscope creates a narrow cone of light and has a small numerical aperture. However, the resolution can be improved with a sub stage condenser. A wide cone of light through the slide and into the objective lens increases the numerical aperture there by improves the resolution of the microscope.

Types of microscopes

In order to view microorganism and microbial structures of different sizes we require different kinds of microscopes.

- Light microscopes resolve images with the help of light. The specimen is viewed as dark object against a light background in bright field microscope. Dark field microscope uses a special condenser and the specimen appears light against a black background. The other types of mircoscopes are Phase contrast and Fluorescence microscope.
- Electron microscope uses a beam of electrons instead of light. Electrons pass through the specimen and form a two dimensional image in Transmission Electron Microscope (TEM). Electrons are reflected from the specimen and produce a three dimensional image in Scanning Electron Microscope (SEM).

2.3 Bright Field Microscope

The most commonly used microscope for general laboratory observations is the standard bright field microscope (Figure 2.6). It contains the following components

- A mirror or an electric illuminator is the light source which is located at the base of the microscope.

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